FH University of Applied Sciences TECHNIKUM WIEN

MASTER’S THESIS

Thesis submitted in fulfillment of the requirements for the degree of Master of Science in Engineering at the University of Applied Sciences Technikum Wien - Degree Program SWE

Project Report Step Quest

By: David Strauß, BSc

Student Number: 52112310

Supervisors: Jürgen Konrad, MSc
Ing. Dr. techn. Dominik Dolezal, MSc

Wien, July 31, 2026

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1 Introduction

1.1 Goal

The goal of this thesis is to design, implement, and evaluate *StepQuest*, a Unity-based mobile exergame that encourages players to integrate regular walking into their everyday routines. The game links real-world step counts to in-game progress by using walking as the central resource for completing daily quests, unlocking content, and advancing through a short narrative structure. Instead of treating physical activity as an external task, StepQuest aims to embed walking directly into the core gameplay loop.

The project addresses the broader problem of insufficient physical activity by exploring how mobile game mechanics can support motivation for low-threshold, everyday movement. Walking was chosen as the target activity because it is accessible, requires no special equipment, and can be performed by a wide range of users in everyday contexts. By focusing on step-based progress, the prototype attempts to create a simple connection between real-world behavior and virtual rewards.

A central goal of StepQuest is to investigate how game design elements such as daily quests, progression, feedback, unlockable content, and narrative framing can be used to support sustained engagement. The design is informed by exergame and gamification research, which suggests that activity-based games need to balance health-related goals with enjoyable and meaningful gameplay experiences [1], [2], [3]. In addition, the project draws on motivational and behavior-change frameworks such as Octalysis and the Behavior Change Wheel to justify the selected mechanics and their intended effects on player motivation [4], [5].

The thesis does not aim to prove the long-term effectiveness of StepQuest as a clinical health intervention. Instead, it evaluates the prototype as an exploratory, theory-informed mobile game and examines whether its design can create motivating experiences around walking. The evaluation focuses on user experience, engagement, motivation, and indications of walking-related behavior during the test period. In this way, the project contributes practical design insights into how mobile exergames can be structured to make everyday physical activity more engaging.

1.2 Background

Insufficient physical activity is a persistent public health problem and remains a relevant challenge for digital health and human-computer interaction research. The World Health Organization identifies physical inactivity as a major risk factor for non-communicable diseases and recommends that adults perform at least 150 to 300 minutes of moderate-intensity aerobic physical activity per week [6], [7]. However, recent global data show that a large share of the adult population does not meet these recommendations. Strain et al. report that the global age-standardised prevalence of insufficient physical activity among adults reached 31.3% in 2022, corresponding to approximately 1.8 billion adults worldwide [8]. This indicates that the problem is not limited to a lack of knowledge about the benefits of physical activity, but also concerns the difficulty of translating such knowledge into regular behavior.

The issue is also relevant in an Austrian context. According to Statistik Austria, only a minority of adults in Austria meet both the aerobic and muscle-strengthening recommendations defined by the WHO [9]. This highlights the need for approaches that lower the barrier to physical activity and can be integrated into everyday routines. Walking is a suitable target activity in this context because it is low-threshold, does not require special equipment, and can be performed in many everyday situations. Research shows that walking interventions can improve several health-related outcomes, including aerobic fitness, blood pressure, and measures related to body composition [10]. Walking-based approaches are therefore well suited for interventions that aim to increase daily movement without requiring users to adopt a demanding sport routine. For a mobile intervention, walking also has the advantage that it can be measured through step counts, making it possible to connect real-world behavior directly to digital feedback and progress.

Mobile technologies offer a promising platform for such interventions because smartphones are already part of many users' everyday lives and can measure physical activity with built-in sensors. Systematic reviews suggest that smartphone applications and activity trackers can increase physical activity, although the observed effects are often modest and may be stronger in the short term than in the long term [11], [12]. This is an important limitation: simply providing tracking or numeric goals may not be enough to create sustained engagement. Digital physical activity interventions therefore need to address not only measurement and feedback, but also motivation and perceived meaning.

Games provide one possible way to address this challenge. By linking physical activity to goals, feedback, rewards, progression, and narrative context, mobile exergames can make movement part of an interactive experience rather than presenting it only as a health-related obligation. Previous research on gamified smartphone applications indicates that gamification can have positive effects on physical activity, but also shows that the effectiveness of such systems depends on the concrete design of the implemented game elements [3]. Similarly, large-scale examples such as *Pokémon Go* demonstrate that location-based and movement-based games can temporarily increase real-world activity, while also showing that novelty effects and declining engagement remain important challenges [13], [14].

This creates the research gap addressed by StepQuest. There is a need for mobile exergame prototypes that do not only count steps, but deliberately structure walking around motivational game mechanics such as daily quests, progression, unlockable content, and narrative feedback. StepQuest explores how these elements can be combined in a Unity-based mobile game to support regular walking and sustained engagement during everyday use. The importance of this research therefore lies in examining how a low-threshold activity such as walking can be connected to game design in a way that is motivating, technically feasible, and suitable for evaluation in a real-world prototype setting.

1.3 Research Questions and Hypotheses

The evaluation focuses on whether StepQuest can provide a motivating and engaging game experience around walking and whether indications of changed walking behavior can be observed during the test period.

Based on this objective, the thesis addresses the following research questions:

- **RQ1:** Can changes in walking behavior and motivation be observed during the use of StepQuest?
- **RQ2:** How do users perceive the motivational and engagement-related design elements of StepQuest, such as daily quests, progression, feedback, unlockable content, and narrative framing?
- **RQ3:** Which aspects of the StepQuest prototype support or limit sustained use during the evaluation period?

These research questions reflect the exploratory nature of the project. StepQuest is evaluated as a theory-informed mobile exergame prototype, with a focus on user experience, motivation, engagement, and walking-related behavior during everyday use. The evaluation therefore combines user feedback with usage-related data collected during the test period.

The following hypotheses guide the evaluation:

- **H1:** Users will report increased motivation to go for walks while using StepQuest compared to before using the prototype.
- **H2:** Users will perceive the step-based quest system as a meaningful connection between real-world walking and in-game progress.
- **H3:** Game elements such as daily quests, progress feedback, unlockables and narrative progression will be perceived as supportive for engagement with the prototype.
- **H4:** Technical or design limitations, such as step tracking reliability, repetition, or limited content variety, will negatively affect sustained engagement.

The hypotheses are based on previous research suggesting that physical activity games and gamified mobile applications can support motivation and activity when they provide clear goals, feedback, progression, and meaningful rewards [1], [2], [3]. At the same time, research on mobile physical activity interventions and movement-based games indicates that long-term engagement is difficult to maintain and that novelty effects can reduce continued use over time [11], [14]. Therefore, the evaluation of StepQuest considers both the motivational potential of its design and the practical limitations that may affect real-world use.

2 Methodology

This chapter describes the methodological approach used for the design, implementation, and evaluation of StepQuest. The project follows a design-oriented research approach in which a functional prototype was developed based on existing exergame, gamification, and behavior change literature. The prototype was then evaluated with users in a real-world test setting to examine motivation, engagement, usability, and indications of walking-related behavior during use.

The methodology consists of four main parts. First, the design rationale and framework selection are explained. This includes the motivation for using walking as the target activity and the selection of Octalysis and the Behaviour Change Wheel as guiding frameworks. Second, the prototype development process is described, including the main gameplay systems and technical implementation. Third, the user testing procedure is presented. Finally, the data collection methods are explained, including questionnaire data, gameplay-related metrics, and qualitative feedback.

2.1 Research Method Selection

The aim of this thesis is to design, implement, and evaluate a functional software artifact. The methodological structure of this thesis follows the Design Science Research Methodology described by Peffers et al. [16]. This process consists of six steps: problem identification and motivation, definition of the objectives for a solution, design and development, demonstration, evaluation, and communication. These steps fit the structure of StepQuest: the problem is insufficient physical activity and the difficulty of sustaining walking motivation; the objective is to design a mobile exergame that connects walking to game progress; the artifact is the implemented Unity prototype; the demonstration takes place through real-world use by participants; the evaluation is based on questionnaires, usage data, and qualitative feedback; and the results are communicated through the analysis and discussion of the thesis.

Prototyping was chosen because the central concept of StepQuest cannot be fully examined on a theoretical level alone. The project combines step tracking, quest progression, unlockable content, narrative feedback, and mobile interaction. These elements need to be implemented in an interactive system before their feasibility and user experience can be meaningfully assessed. In design science research, the creation and evaluation of an artifact is considered a way to generate knowledge about a problem and a possible solution [15], [16]. For StepQuest, the prototype serves as this artifact: it translates the theoretical design assumptions into a concrete mobile game that can be tested by users.

The prototype also makes it possible to examine technical and practical constraints that would not be visible in a purely conceptual design. These include the reliability of step tracking on Android devices, the clarity of the quest flow, the usability of the interface, and the suitability of the daily quest structure for real-world use. Since StepQuest is intended to be used outside a laboratory setting, the prototype needed to be functional enough for participants to install and use it during their normal routines.

User testing was selected as the second main method because the success of StepQuest depends strongly on user perception and interaction. The prototype aims to motivate walking through game mechanics such as daily quests, progress feedback, unlockable content, and narrative framing. Whether these mechanics are understandable, motivating, or frustrating cannot be determined only from the implemented features. It requires feedback from users who interact with the prototype. User testing is therefore appropriate for identifying usability issues, evaluating perceived engagement, and collecting qualitative feedback on the strengths and limitations of the design [17], [18].

2.2 Design Rationale and Framework Selection

Before implementation, StepQuest was designed around established principles from exergame research and behavior change theory. Exergame design literature emphasizes that successful activity-based games must balance exercise-related goals with enjoyable, safe, and sustainable gameplay. In particular, activity-based games should provide clear goals, meaningful feedback, appropriate progression, and avoid mechanics that interfere with the physical activity itself [1], [2]. For StepQuest, this meant that walking should not be treated as an isolated health task, but as a core part of the game loop.

The design goal was to create a low-threshold mobile exergame that encourages regular walking through daily quests, step-based progress, unlockable content, and narrative framing. Walking was selected as the target activity because it is accessible, requires no special equipment, and can be integrated into everyday routines. From a technical perspective, walking also allows progress to be measured through step counts, which makes it suitable for implementation on Android smartphones using built-in motion sensors.

For motivational design, StepQuest uses the Octalysis framework to structure game mechanics according to different motivational drivers [4], [5]. Octalysis describes user motivation through eight core drives, such as Epic Meaning and Calling, Development and Accomplishment, Ownership and Possession, Scarcity and Impatience, Unpredictability and Curiosity, and Social Influence and Relatedness. This perspective is suitable for StepQuest because the game aims to make walking more engaging by connecting physical activity to meaningful in-game goals and rewards. The framework was chosen because it helps translate abstract motivational concepts into concrete design decisions, such as daily quests, step-based progress feedback, unlockable ship parts, and narrative progression.

To complement this game-centered perspective, StepQuest also considers the Behaviour Change Wheel (BCW). The BCW links behavior change intervention functions, such as education, incentivisation, persuasion, and enablement, to determinants of behavior through the COM-B model: capability, opportunity, and motivation [19]. In the context of StepQuest, the BCW is used to justify how selected mechanics may plausibly support real-world walking behavior. This combination allows the design to be discussed from both a game design perspective and a behavior change perspective.

2.2.1 Intended Octalysis and BCW Mapping

Octalysis organizes user motivation into eight core drives, each representing a different reason why a person may engage with a system. *Epic Meaning and Calling* refers to acting in support of a larger purpose. *Development and Accomplishment* concerns progress, mastery, and completing challenges. *Empowerment of Creativity and Feedback* describes motivation through meaningful choices, experimentation, and responses to player actions. *Ownership and Possession* arises from collecting, creating, or improving something that feels personally valuable. *Social Influence and Relatedness* includes social comparison, cooperation, recognition, and a sense of connection to others. *Scarcity and Impatience* is based on limited access to opportunities or rewards. *Unpredictability and Curiosity* motivates users through uncertainty and the desire to discover what happens next. Finally, *Loss and Avoidance* concerns the motivation to prevent the loss of progress, rewards, or opportunities [4], [5].

Before development, the proposed mechanics of StepQuest were mapped to the eight core drives of the Octalysis framework and to relevant intervention functions of the Behaviour Change Wheel (BCW) [4], [5], [19]. The purpose of this mapping was to translate general motivational and behavior-change concepts into concrete design ideas for the prototype.

Epic Meaning and Calling was addressed through a narrative setting in which the player is stranded on an unfamiliar planet and must recover missing ship parts and information. Daily walking challenges were therefore intended to contribute to a larger in-game objective rather than functioning only as isolated step targets. From a BCW perspective, this narrative framing was associated primarily with *Persuasion*, as it aimed to present walking within a positive and meaningful context. Short health-related information was additionally associated with *Education*.

Development and Accomplishment was intended to be supported through clear step goals, progress feedback, quest completion, and visible long-term progression. Within the BCW, these rewards were associated with *Incentivisation*, since successful walking behavior would result in in-game progress and unlockable content.

Empowerment of Creativity and Feedback was considered through the possibility of allowing players to choose between walking goals with different step requirements. This was intended to provide a degree of autonomy and allow players to select a task that suited their current situation. Immediate step feedback would show how real-world movement affected the game state. From a behavior-change perspective, goal selection and feedback were expected to support self-regulation and enablement.

Ownership and Possession was addressed through collectible ship parts and story modules. These items were intended to remain permanently available in a collection and to create a persistent representation of previous accomplishments. Frequent interaction during walking was avoided because continuous smartphone use could distract from the physical activity itself.

Possible mechanics related to *Social Influence and Relatedness* included achievement sharing, trophy displays, and cooperative walking tasks. These ideas were associated with the BCW intervention function *Modelling*, since other users could provide social comparison or examples of the desired behavior.

Scarcity and Impatience was considered through a daily limitation on quest completion. Restricting players to one reward per day was intended to create a regular progression rhythm and encourage repeated use rather than allowing all content to be completed in a single session.

Possible mechanics for *Unpredictability and Curiosity* included random discoveries, mystery rewards, and short questions related to health or the game's narrative. Health-related questions could additionally contribute to the BCW intervention function *Education*.

Finally, *Loss and Avoidance* was considered through time-limited opportunities, such as rewards that would expire after a defined period. Such mechanics could correspond to the BCW intervention function *Coercion*, although the intended implementation was deliberately mild in order to avoid making missed activity feel punitive.

The mapping provided an initial design space rather than a requirement to implement every proposed mechanic. The final selection was determined during prototype development based on feasibility, complexity, safety, and compatibility with the intended walking experience.

2.3 User Study

A user study was conducted to investigate how participants interacted with StepQuest during everyday use. The study focused on perceived walking motivation, engagement with the prototype, usability, and the relationship between real-world walking and in-game progression. Questionnaire responses were combined with usage data recorded by the prototype.

The study was exploratory and did not include a control group. Consequently, it was not designed to establish that StepQuest caused changes in participants' physical activity. Instead, it examined participants' experiences with the prototype and indications of changes in walking behavior and motivation during the study period.

2.3.1 Participants and Testing Procedure

2.3.2 Data Collection

2.3.3 Data Analysis

3 Results

3.1 Prototype Outcome

The development process resulted in a functional Android prototype that implements the central StepQuest gameplay loop. Players can select a walking quest, accumulate progress through real-world steps, complete the quest after reaching the required step target, and unlock ship parts or story-related content. The prototype also preserves progress between sessions, restricts players to one completed quest per day, and records usage-related information for the user study via logging.

3.1.1 Technical Overview

StepQuest was implemented using the Unity game engine and C#. Unity was used for the game logic, user interface, asset management, persistence, and Android build process. The final application was distributed as an Android Package Kit (APK), allowing participants to install the prototype directly on compatible Android smartphones without publication through an application store.

Android was the primary target platform because it provides access to built-in step-counting sensors and allows applications to be distributed directly for testing. The prototype uses the Android sensor system to obtain physical activity data and connects these data to the Unity-based quest system.

Table 1 summarizes the main technologies used in the development of the prototype.

3.1.2 Software Architecture

The prototype was divided into separate systems for step tracking, quest management, persistence, user interface control, content unlocking, and data logging. This separation reduced dependencies between the physical activity sensor and the gameplay logic. For example, the quest system receives step updates through a common step-counter interface rather than directly accessing the Android sensor implementation.

The step-counter interface has separate implementations for Android devices and the Unity editor. On Android, sensor data are obtained from the device's native step sensors. During development in the Unity editor, a mock implementation allows steps to be added manually. This made it possible to test the complete quest flow without performing physical walks during each development iteration.

Table 1: Main technologies used in the StepQuest prototype

Technology	**Purpose**
Git	Version Control
Unity	Game engine, interface implementation, scene and asset management, and Android builds
C#	Implementation of the gameplay logic, quest system, persistence, and logging
Android Sensor API	Access to step counter and step detector sensors
Local device storage	Persistence of active quests, unlocked content, and daily progress
APK distribution	Direct installation of the prototype on participants' Android devices

The central quest state is managed by a dedicated quest manager. This component stores the currently active quest, the number of accumulated steps, and the unlocked content. It also notifies interface components when a quest is selected, progress changes, or a quest is completed.

Quest content is represented through Unity ScriptableObjects. Each quest definition contains configuration data such as a unique identifier, step requirement, displayed text, quest type, and corresponding unlockable content. Separating this information from the gameplay code made it possible to add and modify quests without changing the underlying quest logic.

3.1.3 Step Tracking

Real-world walking progress is measured using the motion sensors available on Android devices. The prototype first attempts to use the Android `TYPE_STEP_COUNTER` sensor. This sensor reports the cumulative number of steps recorded since the device was last restarted. When a quest or tracking session begins, StepQuest stores the current cumulative value as a baseline. Subsequent progress is calculated as the difference between the current sensor value and this baseline.

If the cumulative step counter is unavailable, the prototype can use the `TYPE_STEP_DETECTOR` sensor as a fallback. In this case, individual step events are counted within the current session. Access to the sensors requires the Android physical activity recognition permission, which is requested by the application on the first startup.

The measured steps are forwarded to the quest system and used to update the active quest. The user interface displays the current number of steps relative to the quest target. Once the required value is reached, the quest becomes eligible for completion.

Because sensor availability and behavior can differ between Android devices, step tracking represented an important technical constraint of the prototype. The implementation was designed to support the two common Android step sensor types, but reliable tracking still depended on the hardware and operating-system behavior of the participants' devices.

3.1.4 Quest and Progression System

The central gameplay loop is implemented through a daily quest system. At the beginning of a quest, the player selects one of the available tasks. Each task defines a required number of steps and an associated reward. During the tutorial phase, the repair-droid quest is mandatory because it establishes the narrative and unlocks the remaining quest structure.

After the introductory quest, the player is presented with a selection of available quests. The prototype distinguishes between ship-part quests and story-related quests. Completing a ship-part quest unlocks a visual component of the damaged spacecraft, while completing a story quest unlocks an information module that contributes to the narrative.

Quest progress is shown through a numerical step display and a progress bar. When the step target is reached, the quest can be completed and the corresponding content is added to the player's collection. The collected ship parts remain visible and gradually reconstruct the spacecraft, providing a persistent representation of long-term progress.

The prototype restricts players to one completed quest per calendar day. The date of the most recently completed quest is stored locally and compared with the current date. Once a quest has been completed, no additional quest can be finished until the following day. This system distributes progression across multiple days and prevents the complete game from being finished in a single session.

3.1.5 Persistence and Data Logging

The prototype stores relevant game state locally on the Android device. Persisted data include the active quest, current quest progress, unlocked content, the date of the most recently completed quest, and the starting date of the playthrough. This allows users to close and reopen the application without losing their progress.

A logging system was implemented to collect usage-related information during the user study. Logged events include quest-related interactions, step progress, quest completion, unlocked content, and relevant application state changes. The logs are stored locally and can be exported through a debug interface after the testing period.

Local storage was selected because the prototype did not require user accounts, a remote server, or synchronization between devices. This reduced technical complexity and avoided transferring participant data to an external backend. However, it also meant that the recorded logs had to be exported individually from each participant's device.

3.1.6 User Interface

The final prototype uses a portrait-oriented mobile interface divided into three primary areas: the home screen, the narrative modules screen, and the ship collection screen. Navigation is provided through a tab-based bar at the bottom of the display.

The home screen presents the current quest, step requirement, and progress. It also contains the dialogue and narrative elements used before and after quest completion. The collection screen shows recovered ship parts, where the user can interact with his ship by rotating it to view from all sides. The narrative modules screen shows the unlocked modules, which provide further explanation of the worlds setting via a text based pop up. Locked content is visually distinguished from unlocked content so that players can see both their current progress and the remaining rewards.

Interaction during walking was intentionally minimized. Players can select a quest before beginning a walk and then carry the smartphone in a pocket while steps are recorded. The interface is mainly used before and after the physical activity rather than requiring continuous attention during movement.

The resulting prototype therefore implements the main functional requirements needed for the user study: physical step tracking, daily walking goals, progress feedback, quest completion, content collection, narrative progression, persistence, and usage logging.

3.1.7 Implemented Octalysis and BCW Elements

The final StepQuest prototype implemented selected elements from the intended Octalysis and BCW mapping. The strongest representation was found in *Epic Meaning and Calling*, *Development and Accomplishment*, *Ownership and Possession*, and *Scarcity and Impatience*. Other core drives were implemented only partially or were excluded from the prototype.

Epic Meaning and Calling

The final prototype embedded the walking quests in a narrative about a stranded pilot recovering missing ship parts and information. The onboarding sequence introduced the setting and explained the overall objective. Walking therefore contributed to progress within the story rather than only increasing a step number on screen.

This mechanic represented the BCW intervention function *Persuasion*, as the narrative gave walking a more positive and purposeful framing. Short health-related messages at the beginning of each day also introduced a limited educational element.

Development and Accomplishment

Each quest in the final prototype contained a defined step target. Progress toward this target was displayed through a progress bar, and completing the quest unlocked a ship part or story module. This created both short-term progress through individual quest completion and long-term progress through the reconstruction of the ship and advancement of the narrative.

The resulting rewards represented *Incentivisation* within the BCW. However, initially considered elements such as points, badges, and leaderboards were not included. Progress was instead communicated through quest completion and unlockable content.

Empowerment of Creativity and Feedback

Players could select between available quests with different step requirements, although the introductory repair-droid quest was mandatory. This provided a limited degree of choice while maintaining the intended narrative sequence.

Feedback was implemented through the continuously updated step counter, progress bar, completion messages, and unlocked rewards. The creativity component remained limited because players could not create their own goals or solve quests in substantially different ways.

Ownership and Possession

Ownership was represented through permanently unlocked ship parts and story modules. Completed quests expanded the player's collection, while recovered ship parts became visible on the reconstructed ship. The collection therefore served as a persistent representation of previously completed walking activity.

The prototype did not require interaction during the walk itself. Players could begin a quest, carry the smartphone in their pocket, and return to the interface later to review progress and rewards.

Social Influence and Relatedness

Social features were not implemented. The final prototype did not include achievement sharing, leaderboards, multiplayer functionality, or cooperative quests. These features were excluded because they would have required additional systems for accounts, networking, privacy, and moderation that exceeded the scope of the prototype. Additionally, other non player characters would not have fit the narrative setting of the abandoned region.

Scarcity and Impatience

The final prototype allowed only one quest to be completed per day. This prevented participants from unlocking all content in a single session and created a progression structure distributed across multiple days.

The limitation was displayed as a fixed rule rather than as an unexpected restriction. No limited-time bonus periods or variable reward schedules were implemented.

Unpredictability and Curiosity

Random rewards, mystery boxes, and quiz challenges were not implemented. Introducing such events during walks could have required notifications or smartphone interaction that conflicted with the goal of allowing the phone to remain in the player's pocket.

Curiosity was instead supported indirectly through the gradual unlocking of story modules and the reveal of additional narrative information. This represented only a limited implementation of the corresponding Octalysis core drive.

Loss and Avoidance

The final prototype did not remove progress, expire previously earned rewards, or punish players for missing a day. Although players could complete only one quest per day, an unfinished or missed day did not result in the loss of collected content.

This decision kept the motivational design focused on positive progression and rewards rather than pressure or punishment. Consequently, the BCW intervention function *Coercion* was not represented in the implemented prototype.

Overall, the final implementation focused on a smaller group of mutually supporting mechanics. Narrative purpose, visible progress, collectible rewards, player choice, and daily limitations formed the central motivational structure. Social systems, random events, and loss-based mechanics were excluded to keep the prototype technically manageable and compatible with a low-interaction walking experience.

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AI-Based Tool	Intended Use	Prompt, Source, Page, Paragraph...
DeepL Translate	Translation of an article in English	Source (XXX), Chapter X on page X-X
Chat GPT 4.0	Grammar and Spelling	"Please list issues with spelling and grammar in the following text: ...", Entire document

List of Abbreviations

ABC Alphabet

WWW world wide web

ROFL Rolling on floor laughing